# Astrarium Pro — User Manual

For version 2.3

22 October 2026


- Contents
- 1. Introduction
  - 1.1 Requirements
  - 1.2 First steps
- 2. The screen
  - 2.1 The top bar
  - 2.2 The signature line
  - 2.3 iPhone Duo (foldable)
- 3. Choosing where you are
- 4. Driving the chart
  - 4.1 All-sky and horizon views
  - 4.2 Gestures
  - 4.3 What a tap tells you
- 5. Moving through time
  - 5.1 Steps and playback
  - 5.2 Grids
- 6. What the chart shows
  - 6.1 The quick bar
  - 6.2 The View tab (≡ → View)
  - 6.3 Export as PNG or PDF
  - 6.4 Demo and recording
- 7. Meteor showers
  - 7.1 The list
  - 7.2 How the rate is worked out
- 8. Comets and asteroids
- 9. Events
  - 9.1 On the Lock Screen and in the Dynamic Island (Live Activity)
  - 9.2 Notifications
- 10. The Info tab
  - 10.1 Cloud cover tonight (forecast)
  - 10.2 Widgets and Siri (iPhone and iPad)
- 11. The Pro features
  - 11.1 Satellites
  - 11.2 AR

    - 11.3 Aurora forecast
    - 11.4 The proper-motion time machine
    - 11.5 The deep catalogue (more stars at telescope fields)
    - 11.6 Telling satellites apart
    - 11.7 The solar system in 3D
    - 11.8 Two more kinds of event
- 12. If something looks wrong
- 13. Glossary
- 14. Credits and contact




# Contents


1. Introduction
2. The screen
3. Choosing where you are
4. Driving the chart
5. Moving through time
6. What the chart shows
7. Meteor showers
8. Comets and asteroids
9. Events
10. The Info tab
11. The Pro features
12. If something looks wrong
13. Glossary
14. Credits and contact


# 1. Introduction

**Astrarium Pro** is a planisphere and astronomical simulator for iPhone and iPad. It shows the sky as it is now, but it also lets you drive the clock freely — back into the past, forward into the future — and it puts planetary phases, eclipses, meteor showers and comets on the same chart.

It also does satellites (the space stations, the naked-eye satellites, science satellites, Starlink, the weather satellites and Japan's geostationary fleet), AR (the chart drawn over the live camera) and the aurora forecast (chapter 11). Everything else is the same screen, and the same handling, as the free Astrarium.

Positions come from the JPL DE421 ephemeris, the IAU precession and nutation models and IERS Earth-orientation data, and are accurate enough to plan real observing sessions with.

This manual is written for **subscribers to Astrarium Pro**. Since 2.1, Pro is a subscription inside the same app rather than a separate one. It opens the 3D solar system map, the aurora forecast, the proper-motion time machine, the deep star catalogue (2.34 million stars to 12th magnitude), exoplanet transits and variable stars, every satellite group with a world map, a globe and search, the equidistant (dome master) projection and local sidereal time. For what the app does without a subscription, see the main manual.

# 1.1 Requirements

| | |
|---|---|
| Devices | iPhone / iPad (iOS and iPadOS 17.0 or later) |
| Orientation | Portrait and landscape both supported |
| Price | Astrarium Pro, an in-app subscription. No advertising |
| Network | The chart and all the astronomy run on the device — it works offline |
| Data collection | None. Your site and settings stay on the device |

An internet connection is needed only to **update** four things:

- satellite orbital elements (TLEs, from Celestrak),
- the aurora forecast (NASA DONKI, NOAA OVATION),
- comet and asteroid elements (from the MPC and JPL),
- your current position (this uses location services, not the network). Where Precise Location is switched off, the app asks for it once, for that single fix: an approximate position can be kilometres out, and that changes what is visible near the horizon.

Two more things go over the network, **only when you use them** (new in 2.3):

- tonight's cloud forecast (from MET Norway; section 10.1). The app asks once before it fetches anything, and sends only your site's latitude and longitude rounded to 0.1° (about 10 km);
- the light-pollution overlay on the site map (from the Astrarium home page on GitHub Pages; chapter 3), loaded when shown, like the map (OpenStreetMap) itself.

## 1.2 First steps

1. On launch, the app shows the sky over Tokyo at the present moment.
2. Press 📍 in the top bar and allow the use of your location. Astrarium switches to your position and tells you the nearest registered site and how far away it is. If you would rather not share your location, pick a site from the list or type in coordinates (chapter 3).
3. Outdoors, press 🔴 (night vision). The whole interface turns red so that it does not spoil your dark adaptation.

# 2. The screen

The screen is divided into five bands, from top to bottom.

| Band | Where | What it does |
|---|---|---|
| Top bar | Top | Clock, observing site, location, panel, language, night vision |
| Chart | Middle | The sky itself — driven with your fingers |
| Quick bar | Below the chart | Magnitude slider and the common toggles (two rows) |
| Time bar | Below that | How the sky is shown (all-sky, horizon, AR, aurora) and the grids; moving and playing the clock (three rows) |
| Signature | Bottom | Logo and author, both links |

The three rows of the time bar are, from the top,

1. **All-sky / Horizon / AR / Aurora / Grids** — how the sky is shown
2. **−syn −1d −1h −1m / +1m +1h +1d +syn** — moving the clock
3. **Date button / time / Now / ◀ ■ ▶** — the date and time themselves

Pressing the date button opens the date card: type the year, spin the month and the day. The time field can be typed into.

**The two lower bars fold away**, in either of two ways:

- **Press the triangle (▼)** just above them; press again to bring them back. The arrow says which way things move (▼ folds down, ▲ brings back).
- **Take hold of a bar itself and drag it up or down.** Past halfway it folds; the triangle pulls it back. Take hold between the controls rather than on a button or slider, so that the ordinary controls keep working.

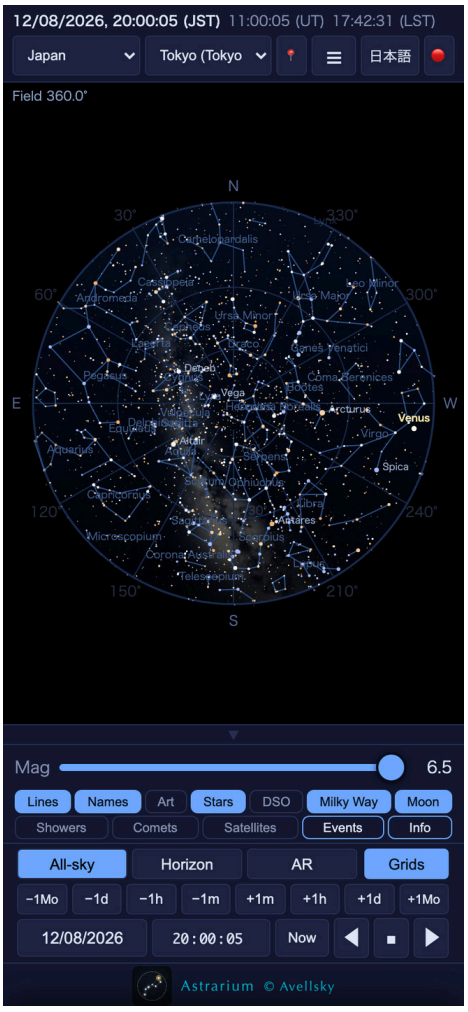



Figure 1. The all-sky view (Tokyo, 12 August 2026, 20:00)

# 2.1 The top bar

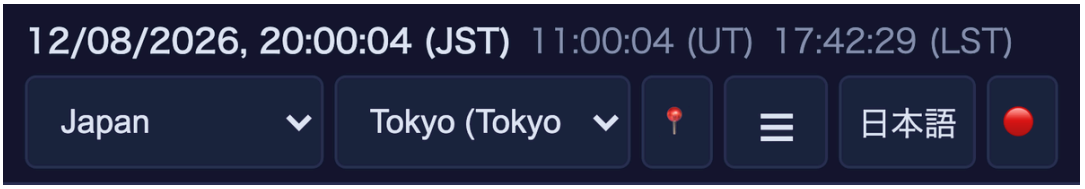



Figure 2. The top bar: site category and site, location, panel, language, night vision

- **Clock** — local time at the observing site and Universal Time (UT), side by side.
- **Site category** — Japan (the 47 prefectural capitals), Japanese dark-sky sites (six more in 2.2, 52 in all), observatories, aurora viewing sites, major cities other than capitals, the capitals of each continent, and the Antarctic research stations. Observatories at home and abroad are one list. Each menu is named for exactly what is in it: “Asia · capitals” holds one national capital per country. Choosing observatories adds a second **Region** menu so you can narrow the list down by continent.
- **Site** — the place itself. The prefectures and the dark-sky sites run north to south; the world cities are alphabetical, by the reading of the name when the app is in Japanese.
- 📍 — use your current position.
- ≡ — open or close the detail panel.
- **EN / 日本語** — switch language. Object names, constellation names and every description change with it.
- 🔴 — night vision; press again to return to normal colours.

## 2.2 The signature line

The logo at the bottom opens the Astrarium Pro home page and `© Avellsky` opens the author’s page, both in the browser.

## 2.3 iPhone Duo (foldable)

On the foldable iPhone Duo the layout follows the way the phone is opened (new in 2.3).

| Pose | Layout |
|---|---|
| Unfolded | The chart spreads across the large inner screen; the panel opens to the side |
| Half open, standing (portrait) | The chart on the upright half; the time bar and panel on the half lying flat |

| Pose | Layout |
| --- | --- |
| Half open, standing (landscape) | The chart on one half, the panel on the other |
| Folded, held sideways | The top bar shrinks to a single row to leave room for the chart; the panel slides in from the right |

# 3. Choosing where you are

Pick a place from the category and site menus in the top bar, or press 📍 to open the **site screen**, which holds a map, coordinate entry and your saved places on one panel.

**Pick it from the list**

Use the category and site menus in the top bar. The app carries the capital of each of the 47 Japanese prefectures — “Sapporo (Hokkaido)”, named with its prefecture — recommended dark-sky sites in Japan ordered by prefecture from the north, every national capital named with its country, the major cities that are not capitals, the Antarctic research stations, and the world’s observatories with their MPC codes. Each place appears in one menu only. **Aurora viewing sites** collects the places people go to see the aurora (23 of them): Fairbanks, Yellowknife, Tromsø, Abisko, Rovaniemi, Akureyri and Longyearbyen in the north, Hobart, Invercargill and Ushuaia in the south, and two in Japan — the Nayoro and Rikubetsu observatories. The bracket after each name is **the country**: Japan comes first, then the countries in order, and the places within a country by name.

**Open the site screen**

Press 📍 . Nothing moves the moment you press it.

- **Get current position** — choose “While Using the App” when iOS asks. If Precise Location is off, the app asks once for the precise fix (an approximate one can be kilometres out, which changes what sits on the horizon). Tick “Get position at launch” and it will do this on every start.
- **Map** — drag the map (OpenStreetMap) until the centre mark is on the spot; + and – zoom.
- **Latitude, longitude, altitude** — type degrees, minutes and seconds, and the altitude; the arrows step one unit at a time. “Use this position” applies it.
- **List** — save up to **ten** places by name and tap a row to go there.

**Show light pollution**

Tick **Show light pollution** under the map on the site screen and an estimate of sky brightness is laid over the map in colour (new in 2.3) — a way to look for dark places. There are seven classes, from the darkest: excellent dark sky (left uncoloured), dark sky, rural, rural/suburban transition, suburban, city / bright suburb, and inner city. Below the map, a line gives the class for the spot under the centre mark with its **Bortle class** and **SQM** (zenith sky brightness in magnitudes per square arcsecond; the larger, the darker) — for example “Sky here: Inner city (Bortle 8-9, SQM ≈ < 18.3)”. It is **an estimate from satellite images of night lights** (NASA Earth Observatory, Black Marble 2016), **not a measured sky brightness**. Terrain, altitude and the night’s air are not accounted for, so near a boundary it can be a class out. The coloured overlay is loaded when shown (it needs a connection); the class under the mark comes from a table inside the app and works offline.

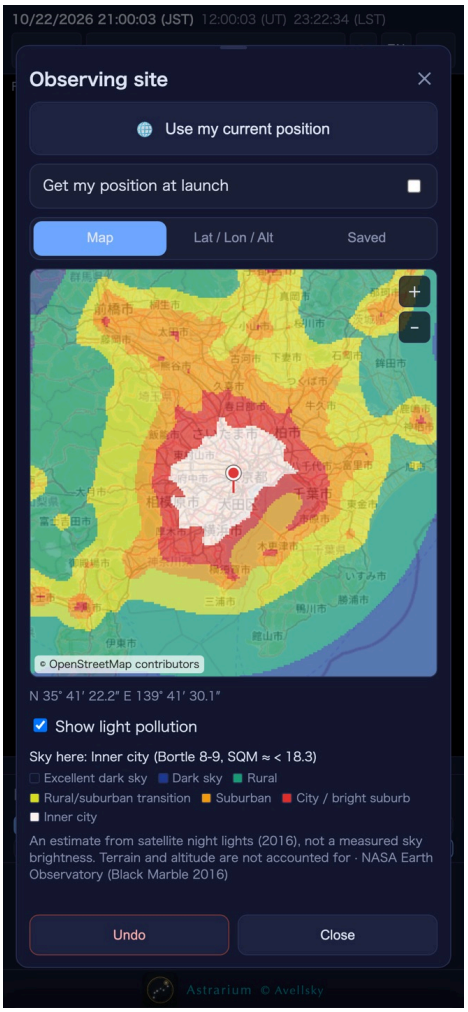



Figure 3. The light-pollution overlay on the site map (Tokyo)

Changing the site updates the clock, the rise and set times and the whole geometry of the chart together.

# 4. Driving the chart

## 4.1 All-sky and horizon views

Switch with **All-sky** and **Horizon** at the left of the time bar.

- **All-sky** — the whole sky as a circle centred on the zenith, north up and east to the left, exactly as on a cardboard planisphere. The Milky Way is drawn in this view.
- **Horizon** — part of the sky drawn as you would actually see it looking out. The menu beside the button chooses which way you face: **N**, **E**, **S**, **W**.

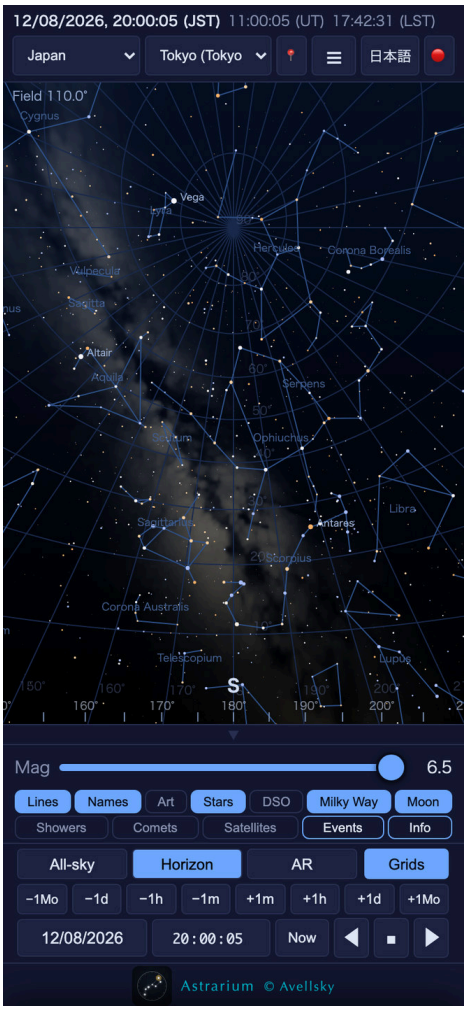



Figure 4. Horizon view, facing south, 110-degree field

## 4.2 Gestures

| Gesture | All-sky | Horizon |
| --- | --- | --- |
| One-finger drag | **Diurnal rotation** — the sky turns about the | Pans the view |

| Gesture | All-sky | Horizon |
|---|---|---|
| | celestial pole and the clock follows your finger | |
| Two-finger pinch | Zoom about the point between your fingers | The same |
| Double tap | Recentre on the zenith | — |
| Tap an object | Opens its information popup | The same |
| Mouse wheel | Zoom | Zoom |
| SHIFT + drag | Pans the view centre without rotating | — |

Dragging in all-sky view is the planisphere gesture: the dial turns and the simulated time turns with it, so you can feel out how high Orion will be in two hours' time with a fingertip.

## 4.3 What a tap tells you

The popup gives the name, right ascension and declination, azimuth and altitude and the magnitude, with buttons underneath. **The readings follow the clock while the card is open.** Azimuth and altitude are recomputed twice a second, $\alpha$ and $\delta$, distance, elongation and phase angle each time the sky is refreshed, and the rise, transit and set times when the local date turns over. "(below the horizon)" is added once the object has set. **A star puts its HD number and its magnitude on the line under the name**, as in "HD 48915 V(-1.5)" — an unnamed star can still be quoted as "HD 143454" and matched against another catalogue or an observing log; having no name is not the same as not being identifiable. Faint stars with no HD number show the magnitude alone.

The Sun, the Moon, the planets and the moons drawn in the zoom view also carry a sentence or two on what they are, and every card gives the **distance** — in astronomical units for the solar system (kilometres for the Moon) and in light years for the stars, from the Hipparcos parallaxes.

Close a card with the **pin (📌) in its top-right corner**. Tapping the body of the card does not close it, so a long description can be read without it vanishing under your finger.

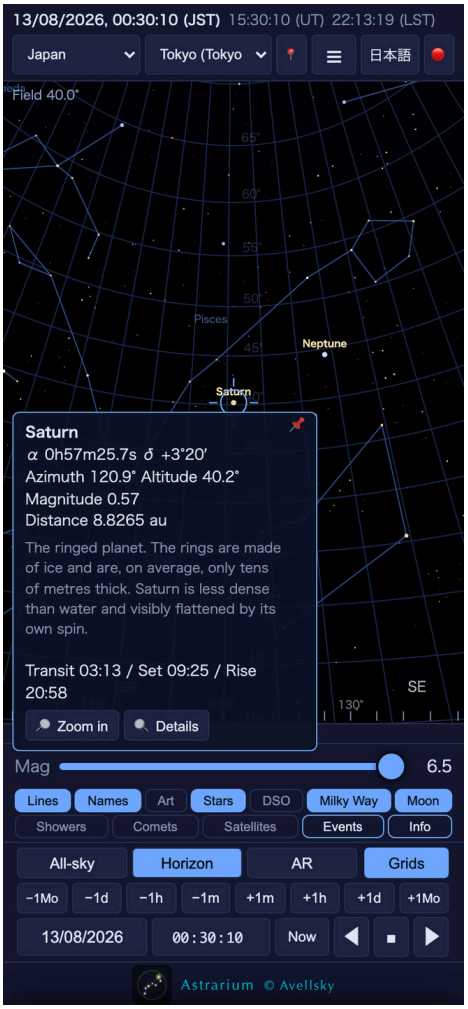



Figure 5. Saturn tapped: coordinates, magnitude, rise and set, and two buttons

- 🔎 **Zoom in** — puts the object in the middle of the screen, zooms in and starts **tracking** it. While tracking, the object stays centred as the clock runs. Release it with the **Release** button on the chart or the Esc key; panning the view does not break the tracking.
- 🔍 **Details** (Sun, Moon and planets) — opens a window that draws the body as a disc, recomputed every five seconds so its distance, elongation, phase angle and libration follow the clock too. The figures include the **physical data** — diameter, mass, mean density, rotation period, axial tilt and surface gravity — and scroll inside the window when there are more than it can show.
- 🧭 **Polar alignment** (Polaris only) — opens the polar scope's field, with Polaris on an hour-angle ring that follows the clock. The red button in that window switches night vision on and off. The Moon comes with its age, illuminated fraction, libration and popular name (Pink Moon, supermoon and so on); Jupiter with the four Galilean moons; Saturn with the rings and Titan; Mars with Phobos and Deimos. You can drag inside the window.

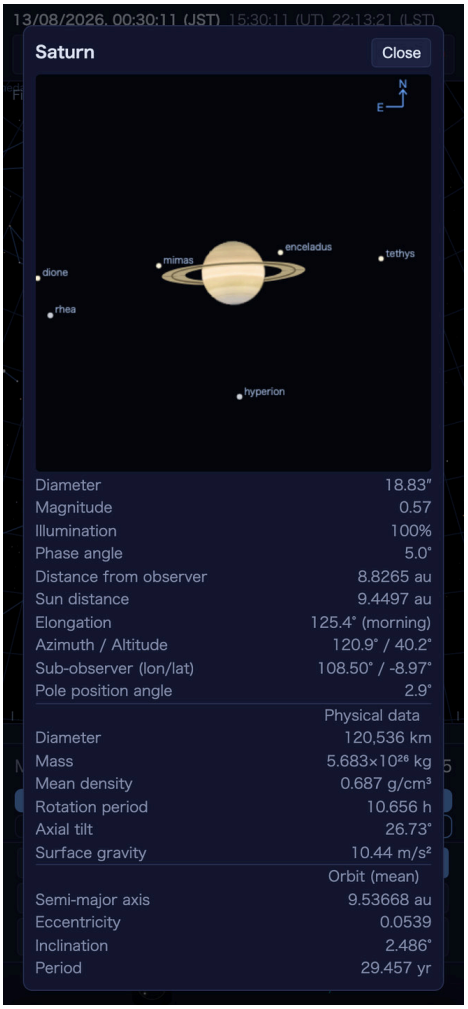



Figure 6. The zoom view of Saturn, with the rings and its moons

Picking a deep-sky object brings up a photograph and a short description; a comet or asteroid brings up its orbital elements and estimated brightness; a satellite brings up its range, height and apparent magnitude with a **Next visible passes** button. Deep-sky names, constellations and descriptions are available in both languages; an object with no Japanese name is listed by its catalogue number in Japanese, with the English name beside it.

**Tapping a meteor shower's radiant** gives its azimuth and altitude as well as its coordinates, and these follow the clock — so you can see at a glance whether the radiant is up yet.

**With constellation art switched on, tapping one of the twelve zodiacal constellations** shows the corresponding astrological sign and its birth dates, together with a note that precession has moved the signs about a month away from where the Sun really is.

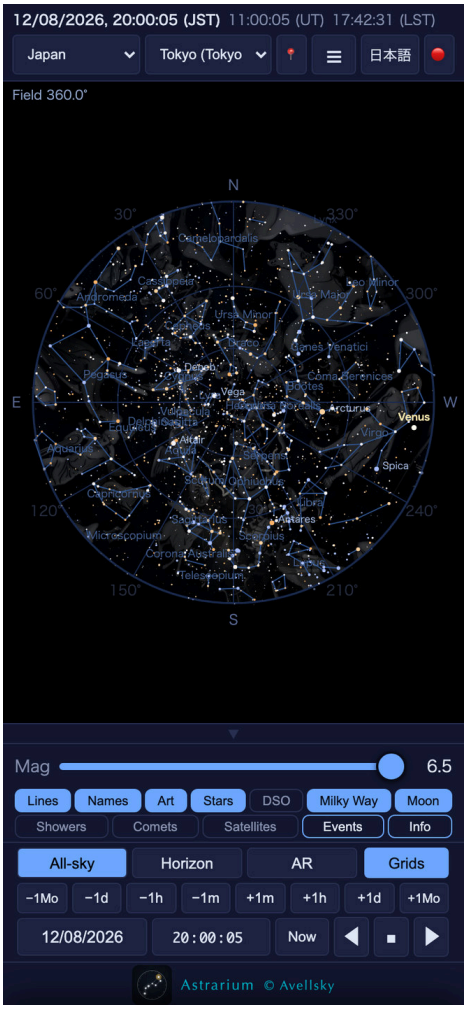



Figure 7. The whole sky with the constellation figures drawn

# 5. Moving through time

The lower two rows of the time bar do this.

## 5.1 Steps and playback

| Control | What it does |
|---|---|
| −syn / +syn | Back or forward one synodic month (29.53 days) — the same lunar phase |
| −1d / −1h / −1m | Back one day, hour or minute |
| +1m / +1h / +1d | Forward one minute, hour or day |
| Date button | Opens the date card. **Type the year; spin the month and day. Go** moves and closes the card; **–** is for BCE. Picking keeps **the time of day** |
| Time field | Type `21:30` and the clock goes to that time on the same day |

| Control | What it does |
|---|---|
| Now | Return to the present |
| ◀ / ▶ | Play backward / forward; each press steps ×1 → ×60 → ×600 → ×6000 and round again |
| ■ | Hold the clock still; press again and it runs from where it stopped |

**The rate is read from the arrowheads**: ▶ ×1, ▶▶ ×60, ▶▶▶ ×600, ▶▶▶▶ ×6000. Running backwards is the same, so ◀◀◀ is ×600 in reverse. At ×600 one second of yours is ten minutes of sky.

**■ also detaches the chart from real time.** Idle, the sky still moves at the rate the real sky moves; ■ stops it dead. Stepping the clock still works while it is held, and it stays wherever you step it to.

With a keyboard, ← and → move ±10 minutes, with `Shift` ±1 day, with `Alt` ±1 minute, and ↑ ↓ move ±1 hour.

**Every date and time is the time at the observing site.** Choose a site abroad and the clock, the date button and the time field all switch to that place's time — not the device's. The zone abbreviation (JST, GMT and so on) follows the clock in brackets.

In the horizon view the altitude lines run all the way to **90° (the zenith)**, numbered like the rest. In the whole-sky view 90° is a point, not a line.

# 5.2 Grids

The **Grids** button opens a menu of overlays; you can have several at once.

| Overlay | What it is |
|---|---|
| Alt-Az grid | Horizontal coordinates — azimuth and altitude |
| RA-Dec grid | Equatorial coordinates, the system star catalogues use |
| Ecliptic grid | Ecliptic coordinates; the planets and the Moon keep to this band |
| Galactic grid | Galactic coordinates; the Milky Way lies along latitude 0° |
| Celestial equator | Drawn as a single great circle |
| Ecliptic | The Sun's path, likewise |

| Overlay | What it is |
| --- | --- |
| Galactic plane | The great circle at galactic latitude 0° — the centre line of the Milky Way |
| Meridian | The north–south great circle objects transit |

The spacing of the lines and the precision of the labels follow the zoom level: roughly **twelve lines across the view**, which over the whole sky means 15° of declination and one hour of right ascension. Zoom in and they tighten; below a degree the labels carry minutes and arcminutes. Where the numbers would crowd, **the lines stay and only the numbers are thinned out** — the count of lines never changes, only how many of them are labelled.

**In the horizon view, switching off the alt-az grid also removes the bearing scale** along the horizon: both say the same thing, so neither is left on its own. The horizon line itself stays — it is the edge of the ground, not a coordinate.

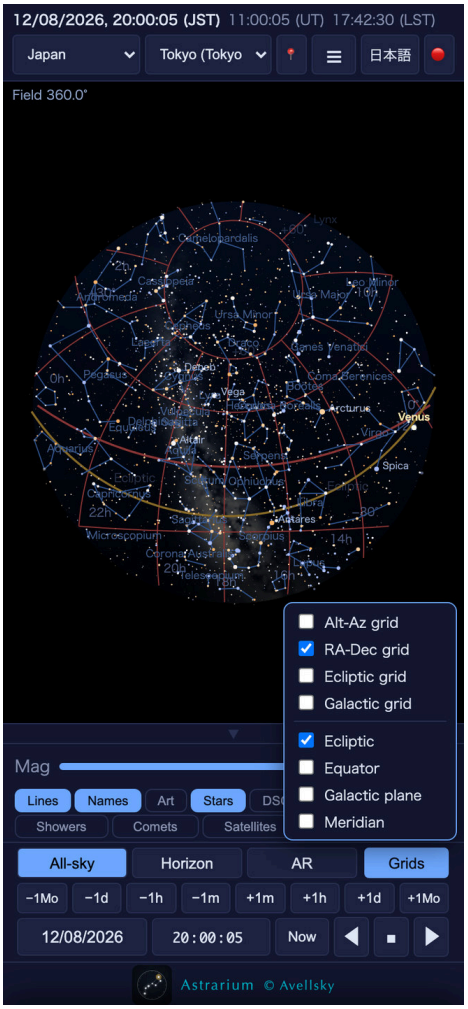



Figure 8. The grid menu, with the RA-Dec grid and the ecliptic drawn

# 6. What the chart shows

# 6.1 The quick bar

The slider on the left is the **magnitude limit**. Lower it and the faint stars go, which is much closer to a city sky; raise it and the chart goes to magnitude 6.5, the naked-eye limit.

The first row controls the chart itself, the second the kinds of object.

| Button | Shows |
|---|---|
| Lines | The figures of the 88 constellations |
| Names | Constellation names |
| Art | Constellation drawings (86 of them; shown whatever the magnitude limit) |
| Stars | Names of the named stars |
| DSO | Messier and Caldwell objects |
| Milky Way | The Milky Way — drawn in the all-sky view, unbroken down to the horizon |
| Moon | Sky brightness from moonlight |
| Showers | Radiants and expected rates of the active meteor showers (chapter 7) |
| Comets | Every comet currently bright enough (fetches elements the first time) |
| Satellites | Satellites on the chart (fetches TLEs the first time; chapter 11) |
| Events | Opens the Events tab |
| Info | Opens the Info tab |

**Comets reach to 12th magnitude, up to 20 of them** (new in 2.2). Only on nights when nothing is brighter than 12th does it go down to 18th, and then for five objects at most. **The count is deliberately not the same every night** — listing the faint ones fills the sky with things you cannot see.

**Cards and panels can be picked up by the strip at the top** (new in 2.2) and moved out of the way.

**AR**, in the first row of the time bar, draws the chart over the camera image, and **Aurora** opens the aurora forecast (chapter 11).

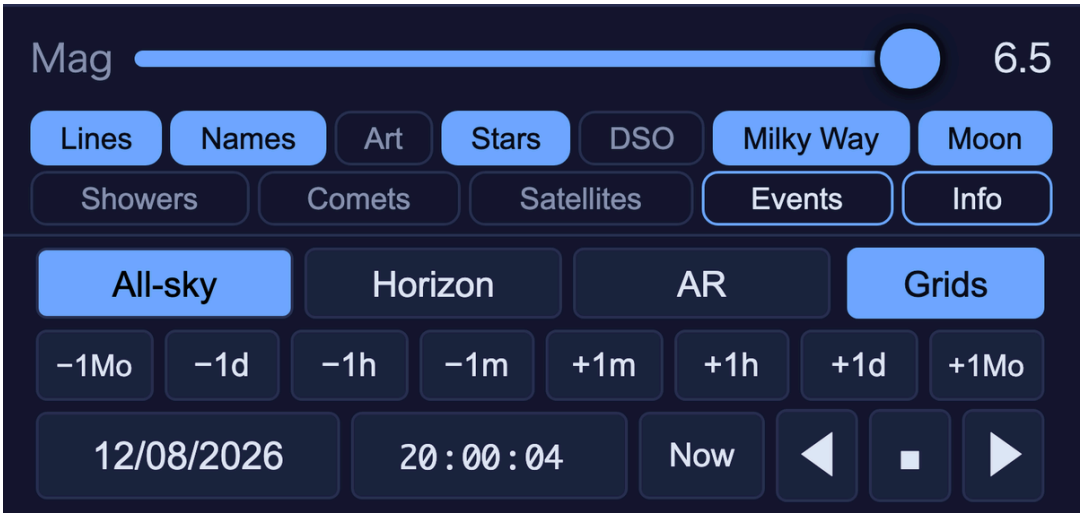



Figure 9. The quick bar and the time bar

## 6.2 The View tab (≡ → View)

Everything the quick bar does not carry lives here.

- **Sun, Moon & planets** — show or hide them.
- **Stars** — show or hide the stars. Switch them off and the constellation lines stand on their own. A star is drawn in the colour of the temperature implied by its own **measured colour index (B–V)**, not a stock colour per spectral class, so two red stars such as Aldebaran and Betelgeuse are not the same red.
- **Sunlight** — daylight and twilight. Switch it off to draw the sky as if it were always dark, which is how you check where the stars are in the daytime.
- **Moonlight** — the wash of a bright Moon. It switches separately from sunlight, so you can leave the daytime sky as it is and take out only the moonlight to see what the stars would do.
- **Mirror** — flips the image for a telescope finder that inverts.
- **Star trails** — leaves a trail behind moving objects during playback, so diurnal motion and satellite passes are drawn as lines.
- **Long exposure** — keeps the trails instead of letting them fade, so the tracks build up as they would on film (switching it on switches trails on too).
- **FOV frames** — rectangles for a camera field or circles for an eyepiece, laid over the chart. “Add frame”, then give it a sensor size and focal length, or an apparent field and a magnification. Drag the centre to move the frame, click the centre to zoom to it, drag a corner to rotate. Useful for framing a photograph, or for seeing what will fit in the field.
- **Mag limit** — the same slider as on the quick bar.

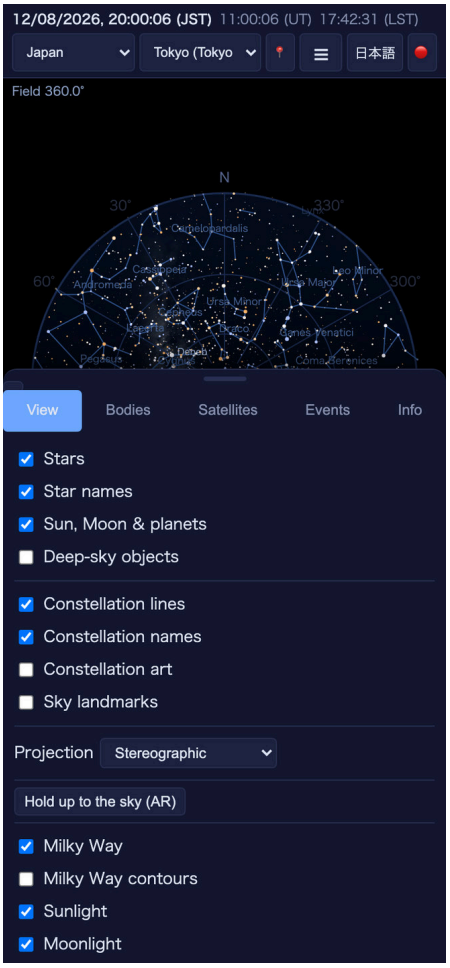



Figure 10. The View tab

# 6.3 Export as PNG or PDF

Saves the chart for the current time and site as an image. Because these are meant to be printed and taken outside:

- the background is white paper and the stars and lines are black,
- sky brightness (sunlight and moonlight) is left out,
- the Milky Way is drawn **as contours only**, not filled,
- the date, time and `Astrarium Pro © Avellsky` appear in the bottom right corner.

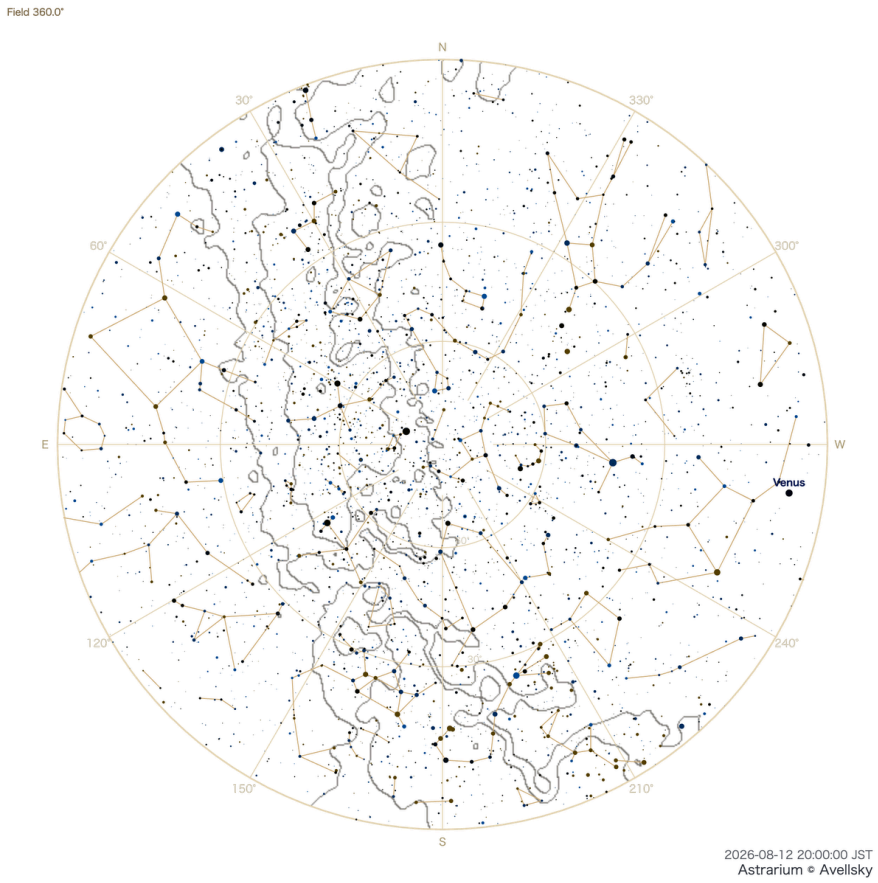



Figure 11. An exported chart: the Milky Way drawn as contours on white paper

## 6.4 Demo and recording

**Demo** is a tour of a few minutes. It plays without sound; each scene is described in captions below the chart. It runs through twilight, the Milky Way and moonlight, the apparent size of a planet, Saturn's rings and moons, FOV frames, the Moon's libration, deep-sky objects with photographs, a meteor shower, a long exposure, the language switch and diurnal motion. Touching the screen ends the tour and restores what you had.

It takes a moment to get ready before the first scene appears — a note under the button says so. While it runs, the button turns red and reads **Stop demo**, so you can end it at any point.

**Record** saves the moving chart as a video (if the device cannot do it, the button says so). Start recording during the demo and the file saves itself when the tour ends.

# 7. Meteor showers

Use the **Bodies** tab or the **Showers** button on the quick bar.

# 7.1 The list

The menu holds the principal showers of the year, sorted by **how close their maximum is to the date on the chart**. Choosing one moves the clock to **the moment of that maximum** and opens the details; where the maximum falls in daylight, it moves to about 1 a.m. that night instead. Clicking the radiant on the chart itself leaves the clock where it is.

| Field | Meaning |
|---|---|
| Maximum | In Japanese, the local time of your site (JST for a site in Japan); in every other language, Universal Time (UT) |
| Radiant | Right ascension and declination, with its altitude at maximum and now |
| Entry speed | How fast the meteoroids meet the atmosphere |
| Population index r | The proportion of bright meteors; the smaller, the brighter |
| Parent body | The comet or asteroid the stream came from |
| Expected rate | Meteors per hour, from the model below |

# 7.2 How the rate is worked out

The figure in “per hour” is not the ideal ZHR — it is **what you can expect from your site at that moment**. Four factors are multiplied together:

1. **Distance from the maximum.** A Gaussian profile: about a day either side the rate has fallen to the sporadic background, and outside the activity period nothing is shown at all.
2. **Radiant altitude.** The lower the radiant, the fewer meteors; below the horizon, none.
3. **Twilight.** Zero between sunrise and sunset, reduced during twilight in proportion to how bright the sky is.
4. **Moonlight.** Reduced according to the Moon’s phase and altitude.

Active showers are also listed in an orange box at the top of the chart. Tapping a name in the box opens its details; **Release** clears the box (and turns off the Showers button on the quick bar with it).

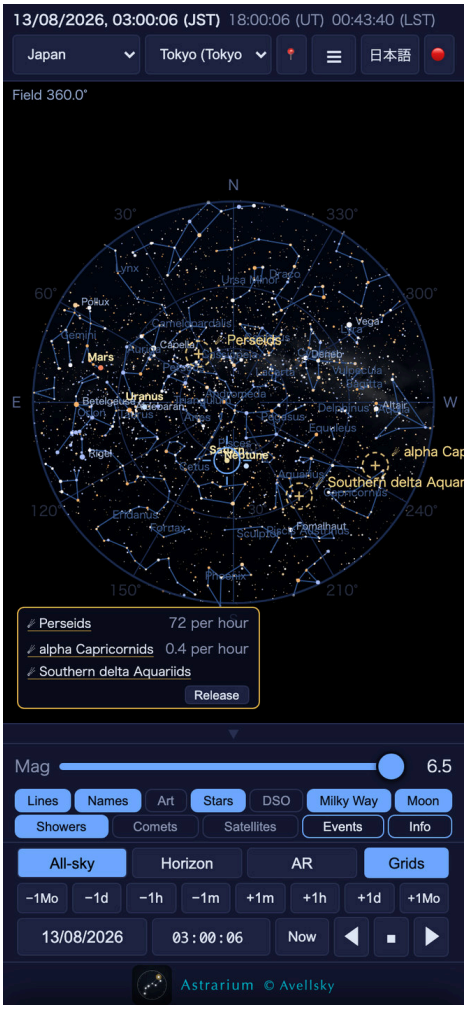



Figure 12. The Perseid maximum (13 August, 3 a.m., 72 per hour expected)

# 8. Comets and asteroids

The lower half of the **Bodies** tab.

- **List bright now** — the comets and asteroids in the sky, brightest first.
- **Search** — by any part of a name or designation. If the object is not in the local data, Astrarium asks JPL for it and adds it (needs a connection).
- **All / Comets / Asteroids** — filters the list.
- **Selected** — the objects drawn on the chart. Anything you put here is plotted.
- **Update from MPC** — fresh comet elements from the Minor Planet Center and asteroid elements from JPL. The date of the last fetch is shown underneath.

Tapping a comet on the chart gives its distance from the Sun and from the Earth, phase angle, estimated diameter, orbital elements, period and epoch.

# 9. Events

The **Events** tab lists eclipses, oppositions and conjunctions, solstices and equinoxes, meteor maxima, close approaches of the Moon and planets, and occultations.

- **Period** — what is next, 1 day, 7 days, 30, 90 or 180 days, 1, 5 or 10 years. “What is next” looks up to 90 days ahead and shows **the next event only**.
- **Direction** — upcoming, past, or both (fixed to upcoming for “what is next”). **Choosing past or both drops the kinds that are only ever predicted forwards** — occultations and satellite passes — from the Kind menu.
- **Kind** — one at a time:
  - **All astronomical events** — eclipses, oppositions, conjunctions and stations, the solstices and equinoxes, the Moon’s phases, perigee and apogee, the meteor maxima.
  - **Occultations** — the Moon passing in front of a star, with immersion or emersion, dark or bright limb, and the star’s magnitude.
  - **Naked-eye satellites** — visible passes of the ISS, the CSS and the Starlink trains.
  - **Eclipses** — with local circumstances (magnitude and times for your site). The last three list only what can actually be observed from your site — above the horizon and in a dark enough sky. Where nothing qualifies, the tab says how many fell in daylight or below the horizon. “All astronomical events” is not filtered.
- 🔍 **Search** starts the scan. While it runs the button reads “⏳ searching 12s — press to stop” and pressing it does stop; the list below is dimmed so that the previous answer is not mistaken for this one.
- Tap a row and **the chart jumps to that moment**; the panel closes so the chart is in front of you.

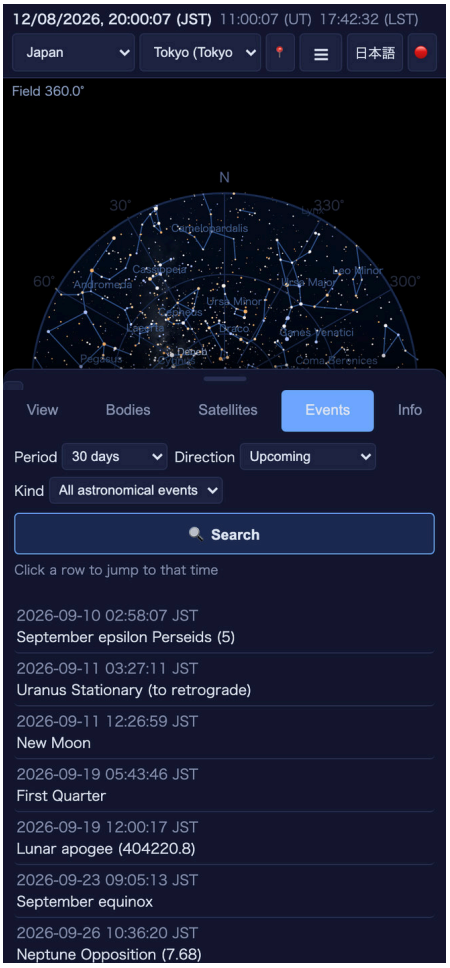


Figure 13. The Events tab

# 9.1 On the Lock Screen and in the Dynamic Island (Live Activity)

Rows for satellite passes and for eclipses visible from your site carry a 📌 on the right (new in 2.3; iPhone only; only for events not yet over). Press it and the event goes onto the **Lock Screen and into the Dynamic Island**: for a pass, the rise time, the highest altitude and the directions it comes from and goes to; for an eclipse, when it begins, its maximum and its end. A clock counts down to the start and then counts up the time elapsed. Tapping it opens the chart at the event's maximum.

The clock is kept by the iPhone, so it runs with the app closed. If there is no 📌, check that Live Activities are on in Settings → Astrarium.

# 9.2 Notifications

Open 🔔 **Notifications** at the bottom of the Events tab to be told ahead of time (new in 2.3; iPhone and iPad only). Tick what you want; the first tick is when the app asks for permission to notify.

| Item | When it is sent |
|---|---|
| Bright ISS pass | **10 minutes** before it rises. Only passes reaching 30° or higher and brighter than magnitude −1.5 |
| Meteor shower peaks | **The evening before at 20:00**, **on the day at 08:00**, and **2 hours before** the peak. Only showers with a ZHR of 15 or more |
| Eclipses visible here | **The evening before at 20:00**, **on the day at 08:00**, and **2 hours before** maximum |
| Full Moon | **2 hours before** |
| Aurora forecast (Pro) | When a forecast brings the aurora into view from your site, about 3 hours before it is due |

- All times are **site times**. If the peak or maximum falls before 11:00, the 08:00 notice is skipped, since it would land right next to the 2-hour one.
- The **site on display** is used. If the schedule is rebuilt while you are looking at somewhere else, the notices follow that place.
- **The schedule is rebuilt when you open the app**: ISS passes a week ahead; showers, eclipses and full moons 45 days ahead. Leave the app closed for long and nothing beyond that is queued.
- The box shows what comes next — “5 scheduled. Next: ⋯”.
- Tapping a notification opens the chart at that event (for a meteor shower, 3 hours before the peak).
- If it says notifications are off, turn them on in Settings → Astrarium → Notifications.

# 10. The Info tab

- **The sky for the date and place you have chosen** — sunset and sunrise, the beginning and end of twilight, the dark window, moonrise, moonset and lunar age, the transits of the Sun and Moon, the traditional lunisolar date and solar term, and local sidereal time. The heading names them both — “Tokyo — 22 August 2026” — and moving the clock to another date changes the table with it.
- **Cloud cover tonight (forecast)** — tonight’s cloud forecast (new in 2.3; next section).

- **Selected object** — the position and rise/set times of whatever you last tapped on the chart.
- **Credits and sources** — every catalogue, ephemeris, image and library the app uses, with its source, its licence and the date it was last checked. It follows the language setting.

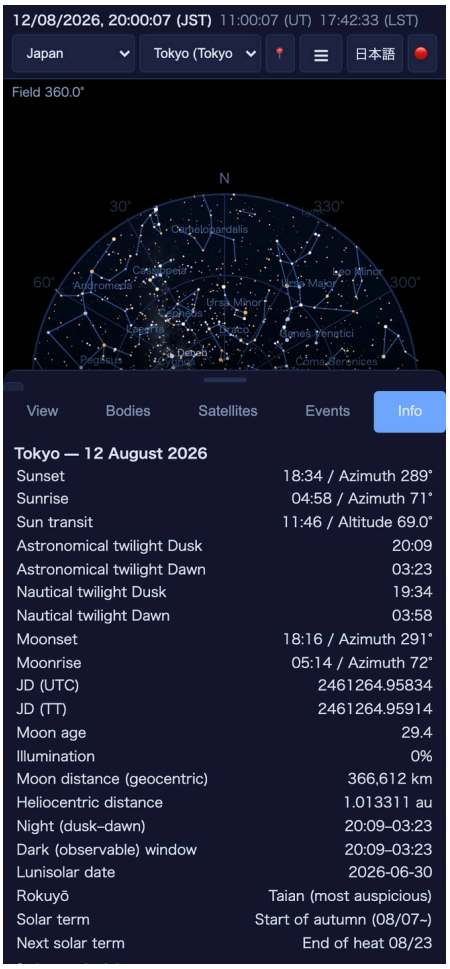



Figure 14. The Info tab, “Tonight’s sky”

# 10.1 Cloud cover tonight (forecast)

Below the table, the Info tab shows **tonight’s cloud forecast** as one bar per hour (new in 2.3).

- The bars run hour by hour from sunset to sunrise, and **their height is clear sky** (the part without cloud): the taller, the better. Times are site times.
- **Outlined bars are the dark hours** (end to start of astronomical twilight).
- The line above gives the average cloud over the dark hours and a verdict: under 25 %, “clear spells likely”; under 60 %, “partly cloudy”; above that, “mostly cloudy”.
- Rest a finger or the pointer on a bar for that hour’s cloud cover, split into low, mid and high cloud.

- Opened in the morning, it shows **the night ahead**, not the one just over.

The forecast comes from **MET Norway** (the Norwegian Meteorological Institute, CC BY 4.0). Unlike the chart, weather can only come from outside, so **the app asks once before fetching anything**: press “Show the cloud forecast” to start. All it sends is your site’s latitude and longitude **rounded to 0.1° (about 10 km)** — nothing else. A forecast is reused for a while; the app fetches at most once an hour.

To stop, press **Stop fetching forecasts** below the box. The forecast is cleared and nothing is fetched until you press “Show the cloud forecast” again.

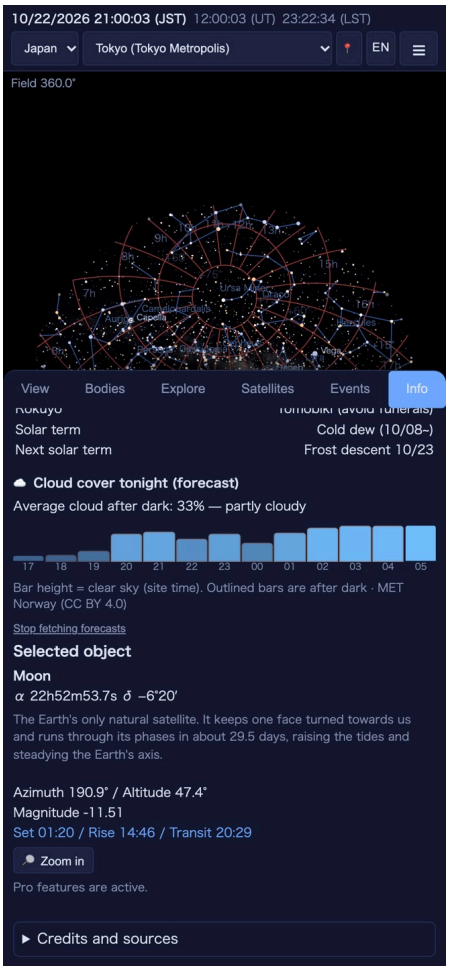



Figure 15. Cloud cover tonight: bar height is clear sky, outlined bars are the dark hours

# 10.2 Widgets and Siri (iPhone and iPad)

The **“Tonight’s Sky” widget** can go on the Home Screen and the Lock Screen (new in 2.3): small, medium and large on the Home Screen; circular, rectangular and inline on the Lock Screen.

| Size | What it shows |
|---|---|
| Small | The Moon’s phase, the dark hours (end to start of astronomical twilight), the next ISS pass |
| Medium | All of that, plus tonight’s cloud bars, the next event (an eclipse visible here, or a meteor shower peak) and the aurora outlook (Pro) |
| Large | All of that, plus the bright passes ahead (up to four) and events (up to three) |
| Lock Screen, circular | The Moon’s phase and illumination |
| Lock Screen, rectangular | The next ISS pass, the dark hours and the average cloud after dark |
| Lock Screen, inline | The next ISS pass (or the dark hours if there is none) |

Tapping the widget opens the chart at the next pass (or now, if there is none).

**Siri and Shortcuts** answer without opening the app.

| Say | Answer |
|---|---|
| “Can I see the ISS in Astrarium” | The next bright pass of the ISS or Tiangong: time, directions, highest altitude, magnitude |
| “Tonight’s sky in Astrarium” | When it gets dark, how much of the Moon is lit, the cloud, the next pass and the next event |
| “Aurora in Astrarium” | Whether a forecast brings the aurora into view from your site (Pro) |

“Next ISS pass in Astrarium”, “What’s up tonight in Astrarium” and “Can I see the aurora in Astrarium” work too, and the same three appear in the Shortcuts app.

**Both show what was worked out when you last opened the app**, because the astronomy runs only inside the app. Once that is more than half a day old, the widget adds **when it was worked out** in orange at the bottom right (“As of Oct 22 21:00”), and Siri says so before answering. Opening the app now and then keeps them fresh. They use the **site on display** at the time.

# 11. The Pro features

These are what the free Astrarium does not have. There, satellites are the space stations only, and there is no aurora forecast and no proper-motion time machine.

**Three of them now sit together under the “Explore” tab in ≡** — the aurora forecast, the solar-system map and the proper-motion time machine, one tab to the right of **Bodies**. They used to be scattered between the bottom of the view tab and a tab of their own; now there is one place to look. Satellites and AR stay where they were (the **Satellites** tab and the time bar).

# 11.1 Satellites

The **Satellites** tab, or the **Satellites** button on the quick bar.

1. Press **Download / update TLEs** first (this needs a connection). Within two hours of a fetch it answers “up to date” rather than asking again. Several groups are fetched at once, so the wait is the slowest one, not the sum. The elapsed seconds are shown while it runs; with no answer after 25 seconds (60 for Starlink) it stops and says so. Starlink is about 1.7 MB and more than ten thousand objects, and propagating them takes another ten seconds or so.

   **About the interval.** CelesTrak asks callers not to fetch the same group more than once every two hours — that is how often the elements change. Fetching harder returns errors, and persisting gets the address blocked. Astrarium Pro records the time of each fetch and, within two hours, answers from what it already has. That is not thrift; it is how the block is avoided.

2. Switch on **Show on chart** and the satellites start moving across the sky.

3. Choose the groups to show.

| Group | What is in it |
|---|---|
| Space stations | The ISS and Tiangong |
| Brightest (visual) | The satellites bright enough to see |
| Science | Research and observation satellites |
| Starlink | The Starlink constellation |
| Weather | Meteorological satellites |

- **Magnitude limit** — 4 by default, which is roughly what the eye can do.
- **Sunlit only** — show only satellites out of the Earth’s shadow, that is, the ones actually visible.

- Tapping a satellite gives its range, height, apparent magnitude and how old its elements are. **Next visible passes** lists the passes for the next five days.
- Above ×600 playback the satellites are paused: the propagation cannot keep up.

## Finding a satellite

Search by any part of the name. Nicknames work too — “Hinode”, “XRISM”, “Daichi”, “Himawari”, “Michibiki” are translated to the catalogue name.

- Press a name and **that satellite is selected**: its coming passes and its **current position** (azimuth and altitude, sub-satellite point, height in km, whether it is sunlit) are shown. Even when there is no visible pass for weeks, where it is right now is still answered.
- **With the horizon view on screen, the selected satellite is brought to the centre** and kept there, so a pass that crosses the sky in minutes is not lost. Panning the chart by hand lets go of it.
- Geostationary satellites (Himawari and the like) are outside the pass prediction. The catalogue entry and NORAD number are shown, with a note.
- **+** adds it to “My list” for next time.

## Projections

The **Projection** menu at the top of the tab shows the same satellites four ways.

| Projection | What you see |
|---|---|
| All-sky | On the zenith-centred chart |
| Horizon | On the chart as you would look up at it |
| World map | Equirectangular, with the ground track behind and ahead |
| Globe | Orthographic Earth, with the day/night terminator |

The map and the globe carry **national borders** (the land ones) as well as the coastline.

They also carry **the sunset line** (thin and solid) and **the end of astronomical twilight** (dotted) (new in 2.3). The solid line is where the Sun is setting now (solar altitude –0.833°); beyond the dotted one the sky is fully dark (–18°). Both are drawn faintly, about as strong as the grid, so that they stay out of the way of the satellites and the auroral oval.

On the map and the globe the names are **placed so that they do not overlap** (where there is no room, the dot is drawn without a name). Selecting a satellite

brings up a card with its **country and flag**, its height and speed, and its next visible pass.




Figure 16. Satellites on the globe

# 11.2 AR

Press **AR** on the time bar and the chart is drawn over the rear camera image. Hold the device up and the constellations line up with the sky in front of you. The first time, iOS asks for camera and motion permission.

The control panel at the bottom has four rows.

| Row | What is on it |
|---|---|
| 1 | **Field of view** and **bearing** sliders, and night mode |
| 2 | What to draw: stars / constellation lines / names / art / landmarks / Milky Way / Sun / Moon and planets / alt-azimuth / RA-Dec / equator / ecliptic |
| 3 | The **magnitude** slider |

| Row | What is on it |
| --- | --- |
| 4 | Date and time, **Now**, **Reset**, the direction you are pointing, the four **shutter** buttons (twilight, Moon, stars, aurora) and **Leave AR** |

- A button brightens when it is on, so what is being drawn is visible at a glance.
- The **bearing** slider corrects the chart when the compass is off. **Reset** throws the correction away and follows the sensor again.
- **Shutter** composites the camera image and the chart into one picture. The four buttons are exposure settings — the darker the subject, the longer. Keep a little of the landscape in frame, or turn the alt-azimuth grid on, and the direction is still readable afterwards.
- **Leave AR** puts back the display settings you had before AR started.

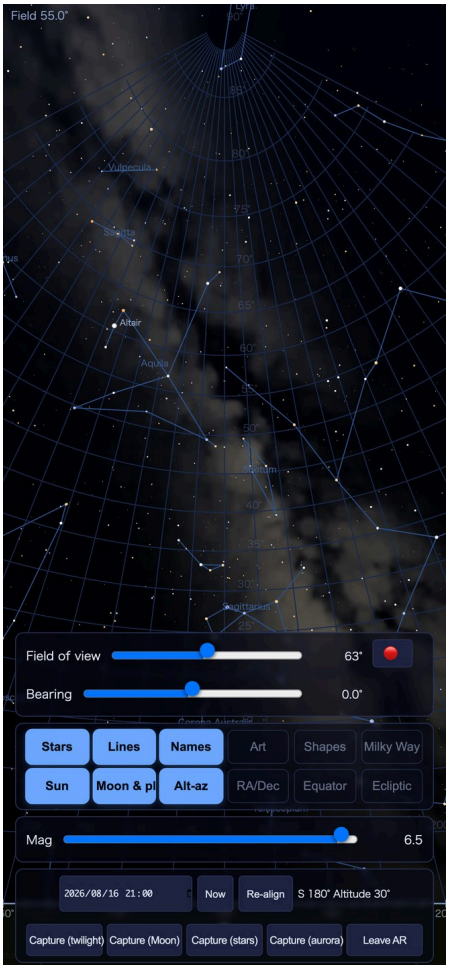



Figure 17. AR: the chart drawn over the camera image

# 11.3 Aurora forecast

**Aurora** on the time bar, or ≡ → **Explore**.

- **Forecast the aurora** fetches NASA DONKI: the solar-wind arrival prediction (the ENLIL model) and the recent geomagnetic storms (observed Kp). Needs a connection. Arrivals within six hours of each other are treated as one storm so the same event is not listed twice; the higher predicted Kp is kept. **The Space weather panel below is refreshed at the same time** (fixed in 2.2: once a copy had been fetched, it could stay there unrefreshed long after it had gone stale).
- The **geomagnetic latitude** of your site is given first, with the Kp at which the aurora would clear your horizon.
- **Draw in the sky** puts the auroral oval on the chart (all-sky, horizon) or on the globe.
- The **Kp** slider lets you see what a stronger storm would look like.
- **Auroral breakup** plays out a substorm (see below).
- Choosing a **projection** and pressing **Auroral breakup** both fold the panel away so the sky is visible. In the horizon view the chart also turns towards **the side the oval is on** — north in the northern hemisphere, south in the southern — at the height the band actually reaches tonight.

## How it is drawn (much changed in 2.2)

The oval is placed by a dipole approximation of the geomagnetic field. From there the colour is **computed from the emission lines themselves** — how strongly each one shines at each altitude, converted to screen colour through the CIE colour matching functions.

| Line | Colour | Where it shines | When |
|---|---|---|---|
| Oxygen 557.7 nm | green | around 115 km | always; the brightest |
| Oxygen 630.0 nm | red | 250–600 km | only high up |
| Molecular nitrogen ion 427.8 nm | blue-violet | 105 km | only in a storm |
| Molecular nitrogen | crimson | 95 km | the foot of a storm |

**There is a reason the red only appears high up.** The state behind the red line lives for 110 seconds, and if it collides with a molecule in that time it gives up its energy without shining. It can only shine where collisions are rare. **A low-latitude aurora is red because three things point the same way: that chemistry, the geometry that hides the low green layer below the horizon, and the optics of an atmosphere that scatters blue-green first.**

Curtains are not vertical. Aurorae are lit by electrons coming down **along the magnetic field lines**, so a curtain leans towards the equator as it rises and its

rays converge on the magnetic zenith. Large folds and vertical rays are drawn as well.

**Pulsating aurora** is there too. From magnetic midnight to dawn, patches tens to a hundred kilometres across **blink with a period of a few seconds**. Only the green pulsates — the red lives too long to keep up. **Stop the clock and they stop.**

The rays along the field lines are **blurred** (fixed in 2.2). A real aurora can throw up sharp rays, but more often it looks softer than that. The curtains are drawn onto a separate sheet and blurred once, at the end, before being laid over the sky.

## Auroral breakup (a substorm)

**Auroral breakup** plays out the moment a quiet arc falls apart and spreads across the whole sky. This is what people travel north to see, and it runs its course in about an hour.

| Phase | Time | What happens |
|---|---|---|
| Onset | seconds | The equatorward-most arc, just before midnight (23h magnetic local time), brightens |
| Expansion | 15 min | It swells 6 degrees (some 670 km) poleward and runs two and a half hours of magnetic local time westward — the westward travelling surge |
| Recovery | 45 min | It fades, and the curtains turn into **blinking patches** |

Inside the bulge **the precipitation hardens**. The electrons are faster and reach deeper, so the emission peak drops: the green deepens and the crimson of molecular nitrogen appears along the lower border. That red-edged foot is a mark of a breakup; a quiet arc does not have it.

**The clock runs at 60x** — an hour is not something to sit through in real time — and goes back to whatever it was doing once the hour is up. Press the button again to stop early. The elapsed phase is shown underneath, as “Expansion phase — 8 min” or “Recovery phase — 31 min”.

**Kp is left alone.** Kp is a planetary index over three hours and sets how far the oval comes down; a substorm is an hour-long event inside that oval, which is a different quantity. Press the button where the oval is below your horizon — most of Japan, for instance — and the elapsed line adds “(below the horizon from here)”.

The aurora forecast is the probability of it being visible **from your site**, based on the NASA DONKI solar-wind model (ENLIL), the observed Kp index and NOAA’s measurements. On the globe, NOAA’s **OVATION** nowcast is laid over it in colour. It is valid for about an hour either side of now; moving the clock returns to the aurora model. The map and the globe also carry the sunset and end-of-twilight lines (section 11.1), so you can read whether the oval lies on the dark side.

**Space weather** (collapsed) lists the background, also from NASA DONKI.

- **Solar wind events** — **interplanetary shocks** caught at Earth (a storm beginning), and **high-speed streams** (wind from a coronal hole; the field tends to stay disturbed for a few days, and the stream returns every 27).
- **NASA/CCMC notifications** — the headlines of the alerts and the weekly summaries, each with a link to the original.

**Neither says whether you would see anything.** Shocks and streams carry no Kp, so no verdict for your site can be drawn from them. What they do tell you, on a night when the two lists above are empty, is whether the Sun is quiet or merely aimed elsewhere.

Below the panel are **Watching and photographing** and **How the aurora works** (mechanism, colour, the solar wind, the magnetosphere, space weather), each with links to its source.

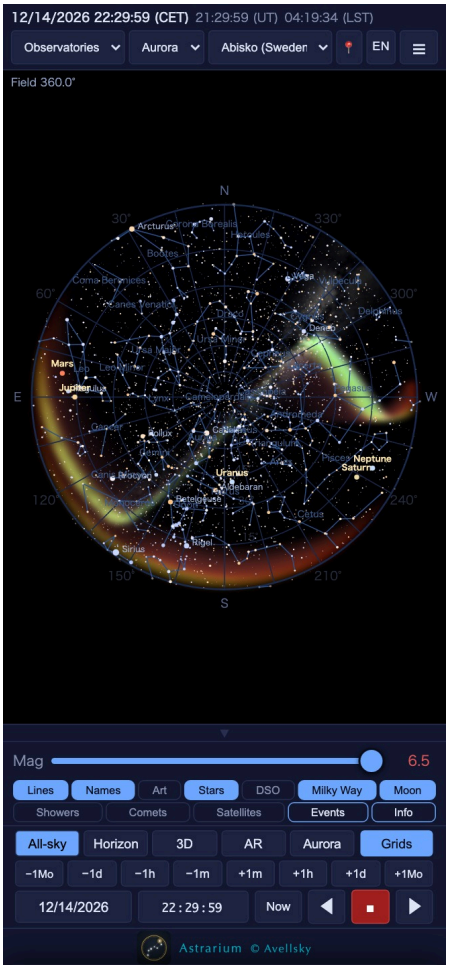



Figure 18. An auroral breakup in progress — ten minutes from onset, the bulge rising poleward and running west

# 11.4 The proper-motion time machine

Every star is moving through the galaxy on its own course. Press **Time machine** on the date card (right of *Now*), or **Open** on the **Explore** tab, and the slider carries the stars along that motion, forward or back, by up to **50,000 years**.

**Time machine** on the date card is a switch: press it to turn the machine on — the button lights — and press it again to turn it off. Turning it on opens the **Explore** tab and scrolls to the slider.

**Type a year and press Go and the panel folds away, leaving that year's sky.** On a phone the panel covers the chart, so leaving it up would hide the very thing you jumped to. Simply moving off the year box sets the year without folding anything.

**Open also switches the sky to the whole-sky view.** What there is to see is the constellations losing their shape, and that only reads on a sky shown whole: the horizon view holds a handful of constellations, and over 50,000 years the ones you were watching swing below the horizon. If the satellite map or globe, the

solar-system map or AR was up, it is folded away and the star chart comes back. If you are already on the whole sky, your projection and zoom are left alone.

**While it is open, the clock at the top and the date on the bottom bar show that year too** (the month, day and time stay as they are). **Back** returns both to the present. The clock itself has not moved: the time machine moves the stars, and the constellation lines and names that hang off them, while the Sun, the Moon and the planets stay where they are.

**While it is on, meteor showers, comets, satellites, Events and Info are unavailable.** A sky moved 50,000 years has none of today's showers, comets or satellites in it: a shower's radiant is set by its parent body's orbit, a comet's elements lose their meaning within decades, and satellites come down within days or years. Events and Info are about tonight's sky too. **Nothing you have chosen is thrown away** — it simply stops being drawn, and comes back when you switch the machine off.

- **Moving the slider computes both the proper motion of the stars (their orbits, with radial velocity and the galaxy's gravity) and the precession of the Earth. The clock stays fixed.**
- Only the stars move, together with the constellation lines and names that hang off them. The clock does not move, so the Sun, the Moon and the planets stay where they are and the sky keeps turning as before.
- You can find, with your finger, the moment the Big Dipper stops being a dipper.
- **Now** restores the present sky, and so does reopening the app.
- **While you drag, only the number moves; the sky is recomputed when you let go.** Solving 5,000 stars is not something to do between two frames of a drag. Typing a year and pressing **Go** moves at once.

## Precession is in it too

It is not only the stars that move. **The Earth's axis wanders** as well (precession), and the slider advances both — so **the celestial pole itself moves**.

- Now: Polaris, $\alpha$ Ursae Minoris
- In about 5,000 years: Alderamin, $\alpha$ Cephei
- In about 12,000 years: **Vega**, $\alpha$ Lyrae
- In about 26,000 years: round again to Polaris

The model is the **long-term precession** of Vondrák et al. (2011), built for ±200,000 years. The ordinary precession polynomials diverge past a few thousand, so they cannot carry a ±50,000-year slider.

**Precession outweighs proper motion here.** Over 5,000 years a star's own motion is under a degree, while precession swings the whole sky by some 70°.

## Type a year

The box under the slider takes a **calendar year**: `1800`, `1800-05`, `-500`. Press Go and the sky moves there.

**Japanese era names come with it** — 1800 reads 寛政12年, 1000 reads 平安時代 (the Heian period).

- Eras are listed from **Keicho (1596)** on; earlier years are answered with the name of the historical period instead. Better to name the period than to print an era we cannot vouch for.
- An era changes partway through its first year. This looks up by year, so **the year of a change reads as the newer era** (1868 is both Keio 4 and Meiji 1; it reads Meiji 1).
- **No era is given for future years.** When the next one begins, and what it will be called, is not decided.
- BCE uses the astronomical convention, which has a year 0: –500 is 501 BCE.

Time of day is not accepted: at the scale of proper motion not even a year is visible (Barnard's star, the fastest, moves 10 arcseconds a year).

## Radial velocity and the Galaxy's gravity are in it

What moves the stars is a **numerical integration inside the Galactic potential** (added 2026-09-21). Both the star and the Sun are carried through the same potential, and the star is then looked at again from wherever the Sun has got to.

- **Radial velocity** — how fast a star approaches or recedes — is taken from the catalogue. **9,090 of the 9,096 BSC5 stars** have a measured value. It matters most for the near ones: including it moves Sirius (2.6 pc, –8 km/s) by about **400 arcseconds** at 10,000 years and **3 degrees** at 50,000. A star coming towards us grows its parallax, and its proper motion speeds up.
- **Galactic gravity** is included. The Sun and the star fall in slightly different directions (a tidal difference), which comes to **0.4–0.8 arcseconds** over 50,000 years. Small — but it is the largest systematic left once radial velocity is in, so fixing only one of the two would be half a job.
- The Galaxy model is a disc, a bulge and a halo, fitted so that **the local gradient of the potential matches the measurements** (circular speed 229 km/s, Oort A = 15.5 and B = –12.5 km/s/kpc, vertical frequency 76 km/s/kpc). Over 50,000 years that is the only part that shows.

- The integrator is time-reversible (leapfrog), so **running forward and back returns the original sky**.

**Beyond 10,000 years it is still an approximation.** What is left is the **parallax error** — the limit of the measurement itself. Fifty thousand years out, a 0.2 % error in Sirius's parallax moves it 10 arcseconds. Star-to-star gravity is not modelled either. **It is one possible arrangement, not a forecast**, and the app says so on screen.

Asterisms (the Spring Triangle, the Summer Triangle and so on) are hidden while the stars are moved: they are traced over today's arrangement, so they would be left hanging in empty sky.

# 11.5 The deep catalogue (more stars at telescope fields)

**Zoom in past a 12° field and stars fainter than the naked eye appear.** The source is **UCAC4** (US Naval Observatory, public domain): 37 MB of tiles holding 2.34 million stars down to V < 12.

**From 2.2 it is downloaded rather than bundled.** Open ≡ → **Explore** and the Deep-sky catalogue panel is at the foot of the tab. **Once downloaded it stays on the device and needs no connection afterwards.** Everyone who does not want it is 37 MB lighter.

| Field width | Limiting magnitude |
|---|---|
| wider than 12° | not drawn (6.5 as usual) |
| down to 12° | 9 |
| down to 8° | 10 |
| down to 3° | 11 |
| down to 1° | 12 |

**Nothing is drawn at wide fields because nothing could be read.** V < 12 is 57 stars per square degree; over the whole sky that is a grey haze. Zooming in is itself the key that opens this layer.

Only what is needed is read. The sky is cut into HEALPix cells (about 0.84 square degrees) and only the cells in view are fetched — move the field and a few new ones follow, a kilobyte each. No connection is required; it all happens on the device.

Faint stars can be tapped like any other, and give their UCAC4 number and magnitude.

**Below a 12-degree field the stars carry their magnitudes** in the form `V(9.4)/R(9.2)`. Where the numbers would crowd they are thinned out, so they stay readable. The magnitude slider also stretches to 12.0 here, so you can choose how faint to go.

**V is measured; R is an estimate.** What is bundled carries APASS Johnson V and B (and for the bright stars, BSC5's V and B-V); there is no R band. It is computed as `R ~ V - 0.5 (B-V)`.

# 11.6 Telling satellites apart

**Colour carries the kind.** The satellite list on the Sky tab shows the same colours as the chart.

| Colour | Kind |
|---|---|
| Green | Space stations (ISS, Tiangong) |
| Gold | Brightest (visual) |
| Blue | Science |
| Violet | Weather |
| Blue-grey, small | Starlink |
| Orange | **Just-launched Starlink** (raising orbit) |

**Starlink markers are deliberately small.** There are ten thousand of them in the catalogue; at the size of the others they fill the sky. Their names are not drawn either — tap one and it tells you.

**Just-launched Starlink stands out on purpose.** A fresh batch goes into a low orbit around 340 km in a train, and climbs to its operating altitude over some weeks. While it does, it is bright and moves in a line — the most conspicuous thing in the sky.

The test is "low orbit **and** from a recent launch". Altitude alone is not enough: Starlink also lowers its orbit at end of life to burn up, so the low orbits hold **satellites on the way up and on the way down at the same time**.

## The magnitude limit uses the best moment of the pass

The magnitude limit in the satellite panel is judged on **how bright the satellite gets at its best during the pass**, not on how bright it is right now.

A satellite just clearing the eastern horizon is far away and faint, and gains several magnitudes by the time it is overhead. Cut on the present brightness and a limit of 4 drops something that is “5th magnitude now”, only for it to appear out of nowhere a few minutes later — which reads as flickering. Judged on the peak, it **stays listed from rise to set**.

The list writes it as “5.6m → 3.5m”: the arrow says it is still brightening.

**Night vision (🔴) keeps one colour.** The red-only mode exists to protect dark adaptation; adding colour back would defeat it.

### The globe shows altitude

On the globe each satellite is lifted off the surface by its altitude. The low-orbit swarm becomes a shell clinging to the ball; geostationary satellites hang well outside the limb. The kind of orbit is legible from the picture alone. The thin line marks the point on the ground directly below.

Up to 2000 km the height is to scale. **Above that it is compressed** — geostationary orbit is 5.6 Earth radii up, and at true scale it would fall outside the screen. The order of heights is preserved.

## 11.7 The solar system in 3D

The Sun at the centre, the planets and their orbits placed in space.

**Two ways in: 3D** on the bottom bar (right of *Horizon*), or **Open** on the **Explore** tab.

**Open rearranges the screen.** The side panel folds away — the map wants the whole of it — and the bottom bar becomes the map’s own.

| The bottom bar | |
|---|---|
| **Back** | Close the map. The panel comes back as it was |
| **Reset view** | Straight down on the plane again |
| two sliders | how far you look down, and the zoom |
| five switches | stars and Milky Way / names / figures / Sun and planets / their names |
| **Small bodies and dwarf planets** | a card: how many to show, and each group |

| The bottom bar | |
|---|---|
| **Spacecraft** | a card: the 35 craft, the search box, the bulk download |

**You move it with your fingers (or the mouse).**

- **Drag** to turn: left and right swing round, up and down change how far you look down on the plane. This is how the **inclinations** show.
- **Pinch or scroll** to zoom.
- “Reset the view” returns to looking straight down.

**The Sun always stays at the centre.** There is no panning: once the centre of a solar-system map drifts, it stops being clear what you are looking at from where. Turning and closing in are the only two moves.

The sliders are still there: when you want exactly face-on, a number is quicker than a finger.

## Choosing what to show

The switches run in the same order as the sky chart’s.

| | |
|---|---|
| **Stars and the Milky Way** | the background sky (6,331 stars to magnitude 6.2, and the Milky Way map) |
| **Constellation names** | names on the background — they say which way you are facing |
| **Constellation figures** | the artwork, laid faintly behind |
| **Sun and planets** | the Sun, the planets and their orbits |
| **Sun and planet names** | the labels only (the bodies stay) |
| **Small bodies** | asteroids and comets |

**Names and figures start switched off.** Laid over a solar-system map they put the orbits and the small bodies underneath them; switch them on when you want to know which way the view is pointing.

## Stars and the Milky Way behind

The background carries the stars **down to magnitude 6.2 (6,331 of them)**, the constellation lines, and the Milky Way — **the same density map the sky chart uses** (from NASA’s Deep Star Maps).

**It is drawn in perspective, so the sky fills the frame** — you are inside the solar system looking out. The background turns with the view: seen from here

the sky is infinitely far away, so it is placed by direction rather than by position. Turn towards the galactic centre and the star clouds and dust lanes come up in front of you.

**The stars follow Meteorium**, the same author's meteor application: the same magnitude limit, the same star sizes and brightnesses, the same line colour and subdivision, the same colour for the names.

**The Milky Way is the chart's own density map.** The ESO photograph was laid here for a while, but it is bright enough to drown the orbits and the small bodies in front of it, so it was put back.

## Small bodies

The bundled small bodies are drawn **coloured by dynamical group**.

| | |
|---|---|
| Red, orange, pink, violet | Near-Earth families (Apollo, Amor, Aten, Atira) |
| Three blue-greys | Main belt: inner, middle, outer (split at the 3:1 and 5:2 Kirkwood gaps) |
| Yellow | Hildas (3:2 with Jupiter) |
| Green | Jupiter Trojans (1:1 — they gather 60° ahead of and behind Jupiter) |
| Cyan | Centaurs |
| Pink | Plutinos (3:2 with Neptune — Pluto's companions) |
| Mauve | Classical Kuiper belt (cubewanos) |
| Light violet | Twotinos (2:1 with Neptune) |
| Deep violet | Scattered disc (perihelion within Neptune's reach) |
| Sky blue | Detached objects (Sedna and the like) |
| **White** | **Dwarf planets** (Ceres, Pluto, Eris, Haumea, Makemake) |
| **Pale amber** | **Dwarf-planet candidates** (Quaoar, Sedna, Orcus — 27 of them) |
| Teal | Comets |

**The trans-Neptunian objects are split four ways by dynamics.** As one group, the bodies that keep step with Neptune (the plutinos), the ones it threw outward (the scattered disc) and the ones it cannot reach (the detached objects)

all lay in one colour, though they were born in different places and live in different ones.

**Dwarf planets are drawn larger and named.** There are only 5 and 27 of them, but they are the best-known bodies on the map; at one pixel you could not tell they were there. Their names follow the “Sun and planet names” switch.

**Tap the Sun, a planet or a dwarf planet for a card.** The dot or the name will do. You get a photograph, how far from the Sun it is at that moment, the orbital period for a planet, and the opening of its Wikipedia article — the same card the spacecraft have.

**Small bodies can be tapped too.** A dot in the belt gives you its designation (“(433)”, “2020 CD3”), its group, how far from the Sun it is now, a/e/i, perihelion and aphelion, and the orbital period. The designation is read **for that one body** when you tap it, not as a table of 1.56 million. A designation that cannot be decoded is left unnamed: better the group and the orbit than a plausible-looking invention.

**Temporarily captured objects (TCOs) are in.** These are small bodies caught by the Earth’s gravity for months or years — “second moons” — before they return to their own orbit around the Sun: 2006 RH120, 2020 CD3, 1991 VG, 2022 NX1 and 2024 PT5. They are drawn larger, like the dwarf planets. No orbit tells you which they are (seen from the Sun they all look like Earth-crossing near-Earth asteroids), so they are picked out by designation.

**The panel keeps the small bodies and the spacecraft apart.** The spacecraft are listed **in launch order** and scroll within their own box — there are 35 of them, and without a box they fill the panel. Tap a name for that craft’s card.

**The boundaries and colours match Asterion**, the same author’s other application, so that one solar system does not need to be read two ways. Any group can be switched off in the legend.

**The whole MPC catalogue is bundled — 1,553,211 asteroids.** The source is MPCORB (the Minor Planet Center Orbit Database): everything from the numbered objects down to the recent one-opposition ones, wherever the MPC judges a reasonable prediction possible.

**Comets went from 951 to 4,122 in 2.2.** The MPC’s comet file is merged with JPL’s SBDB, which brings in the one-time visitors and the sungrazers. Where both hold the same comet the fresher orbital elements win, so nothing appears twice.

**Comets are coloured by dynamical family** (new in 2.2), split on the Tisserand parameter with respect to Jupiter.

| Family | What it is |
|---|---|
| Jupiter-family | short-period comets governed by Jupiter |
| Halley-type | periods of 20–200 years; steeply inclined, some retrograde |
| Encke-type | stay inside Jupiter's orbit |
| Saturn / Uranus / Neptune family | governed by that planet |
| Kreutz sungrazers (subgroups I and II) | fragments of one great comet |
| Long-period | near-parabolic; from the Oort cloud |

Comets that no longer exist (designations beginning D/) are not drawn.

**The slider (0–100 %) is the share of the catalogue that is drawn, and 100 % is all of it.** The row under the slider reads "drawn / carried".

Drawing all 1.5 million costs about **a second to solve and a tenth to plot** on the development machine (it is solved again whenever the clock moves), and several times that on a phone. If it feels heavy, pull the slider down — 10 % is still 150,000 points, enough to read the belt's texture and the Kirkwood gaps.

Every group is taken brightest-first, so thinning leaves a clean cut: everything above some magnitude is present. Small groups — the Atiras are 39 objects — are never divided by the budget, or they would vanish.

**Fast-forward thins them automatically.** Standing still, density is the point (the belt's texture, the Kirkwood gaps); moving, the motion is the point, and every frame has to be solved again.

**The small bodies move with the clock too.** What arrives is orbital elements rather than positions, and the display solves Kepler's equation for each moment. Fast-forward and you watch the belt turn faster on the inside than on the outside — Kepler's third law, straight off the screen.

**This map runs on a ladder of its own.** At ×1 the planets look frozen, so it starts at a day per second and climbs to a week, a month and half a year per second. The arrowheads read as they do on the chart: ▶ is a day per second, ▶▶▶▶ half a year.

## The planets and the spacecraft move too

**A planet slides along the ellipse you can see.** It arrives as orbital elements rather than a position and is solved like any small body; the orbit line comes from the same state, so a planet never drifts off its own line.

It is a two-body solution, so the further the clock goes the further it departs from the truth (measured: 0.04 au for the Earth ten years out — about two pixels here). Once the clock is half a year from the anchor, the planets alone are fetched again and re-anchored.

**The bundled ephemeris (DE421) covers 1899–2053.** Outside that there is nothing to re-anchor against, so the map keeps going two-body from the last anchor: better slightly wrong and moving than stopped.

**Spacecraft follow the path they actually flew**, sampled from Horizons and read off by date — **including ahead of now**: an active craft is fetched three years into the future (or as far as Horizons goes). A finished flight stops at its last point (Cassini in 2017, Rosetta in 2016).

**From 2.2 those paths are carried inside the app** — 35 craft, 370,000 points. Voyager's and Cassini's tracks are drawn where there is no signal, and the sampling is fine enough that no orbit looks like a polyline.

## New ways to work the map in 2.2

- **Tap a body and its orbit is drawn.** Asteroid or comet, you can see at a glance whether it belongs to the belt or to the outer system.
- **Two fingers pan the map; pinch zooms.** The Sun no longer has to sit in the middle.
- **Bodies you search for and add move along their orbits as the clock runs.**
- **Cards and panels can be picked up by the strip at the top** and moved out of the way.
- **A Data panel** (see "Fetching it all in advance" below) says where each table came from and when, and can fetch it again or go back to the one shipped with the app.

## How time moves here

**While the map is open, one day is the smallest step.** A minute or an hour moves nothing at this scale (0.04° for the Earth, 0.003° for Jupiter), so those buttons would do nothing. Fast-forward starts at a day per second and climbs by week, month and half-year. Back on the sky chart, minutes and hours behave as before.

## Fetching it all in advance

**Download from JPL (SBDB)** fetches the paths of all 17 spacecraft, plus any small bodies you have added, and keeps them on the device. **The next time you open the map no connection is needed** — the button is for pressing before you go somewhere without a signal.

It reports how many were kept. Device storage has a limit, so if some did not fit it says how many. Normally there is no need to press it: a spacecraft is fetched when you tick it.

## What you can add from JPL

**Spacecraft** (tick one and it appears — **35 of them**)

The list runs **in order of launch**, from Mariner 2 (1962) to Europa Clipper (2024): read it top to bottom and it is the history of solar system exploration. The launch year is shown beside each name.

| | |
|---|---|
| 1960s–70s | Mariner 2, Pioneer 10 and 11, Voyager 1 and 2 |
| 1980s–90s | Galileo, Ulysses, NEAR Shoemaker, Cassini, Nozomi, Stardust |
| 2000s | Genesis, Hayabusa, Rosetta, MESSENGER, Deep Impact, New Horizons, Dawn |
| 2010s | Akatsuki, Juno, MAVEN, Hayabusa2, OSIRIS-REx, Parker, BepiColombo |
| 2020s | Solar Orbiter, Perseverance, Lucy, DART, James Webb, EQUULEUS, JUICE, Psyche, Hera, Europa Clipper |

**Press the name, or the dot on the map, and the spacecraft describes itself**: a photograph, the launch date, and the span over which its path can be drawn. The text is the opening of the Wikipedia article (CC BY-SA 4.0) and the photograph comes from Wikimedia Commons — free licences only. The attribution is printed on the card.

**The span is not the same as the launch.** It is what JPL Horizons holds:

- **Hayabusa** begins only in January 2009. The cruise from the 2003 launch out to Itokawa and back is not published, so what can be drawn is the final leg home. Add Itokawa as a small body (25143) and the journey still reads.

- **Rosetta** ends in October 2016 (it landed on the comet) and **Cassini** in September 2017 (it was flown into Saturn).
- Ulysses, Dawn, DART, NEAR and Perseverance are cut off here even though Horizons goes further: the spacecraft no longer exists, and drawing a prediction of where the wreckage would be as “where it is now” would be a lie.

**Viking and Magellan cannot be shown.** They are not in the Horizons spacecraft list (their names return asteroids instead). Of the 1970s missions, only the Pioneers, the Voyagers and Mariner 2 still have an ephemeris there.

## 11.8 Two more kinds of event

The Events tab holds two more kinds (the range and direction are the same as in the free version).

- **Exoplanet transits** — the transits observable from your site.
- **Variable stars** — predicted maxima and minima of the main variables.

Pressing a row moves the clock to that moment and **marks the target** on the chart.

The mark is drawn in either view, but **the whole-sky view is left where it is**. Its centre is the zenith and the horizon is the circle around the whole chart; pulling the star to the middle would move the centre of the projection and push that circle off to one side, changing the shape of the sky itself. **Switch to the horizon view and the chart turns to face the star.** In the horizon view it turns as soon as you press the row.

# 12. If something looks wrong

**Location does not work**
: Settings → Privacy & Security → Location Services, and set Astrarium to “While Using the App”. You can also skip location altogether and type coordinates into the Info tab.

**The event list is slow**
: Only the first 5- or 10-year scan, which takes a few minutes. It is cached afterwards.

**The aurora forecast will not load**
: It needs a connection. Without one, the Kp slider still shows what a storm would look like — that part is computed on the device. **No comets appear**

The Comets button shows nothing when no comet is brighter than the limit, and fetching current elements needs a connection.

**No meteor rate is shown**

You are more than a day from the maximum, or outside the activity period. The rate is also zero in daylight.

**Too few or too many stars**

Move the magnitude-limit slider. To match a real sky, a city is about 4, the suburbs 5-something, and a dark mountain site 6.5.

**No satellites appear**

Either the orbital elements (TLEs) have never been downloaded, or playback is faster than ×600. With “Sunlit only” ticked, craft in the Earth’s shadow are hidden.

**TLEs will not download, or the address has been blocked**

CelesTrak answers a caller that fetches the same group more often than every two hours with an error, and blocks the **IP address** if it continues. The block is against the connection, not against your device or Apple ID, so:

1. Stop fetching and leave it alone for a while. Astrarium Pro does not touch the network within two hours of a fetch, so ordinary use will not get you blocked.
2. Switching networks (Wi-Fi to mobile data, or the other way) changes the address and often works. On shared Wi-Fi the cause may be someone else on the same connection.
3. If it does not clear, ask CelesTrak (celestrak.org) to lift it.

While blocked, **the elements you have already fetched still work**. They lose accuracy over a few days, but the chart and the pass predictions keep running without a connection.

**The event list is slow**

Only the first 5- or 10-year scan, which takes a few minutes. It runs in the background and fills in when it finishes; the result is cached, so the next one is immediate.

**In AR the chart does not line up with the sky**

Steel-framed buildings and cars upset the compass. Step outside and press **Reset**, or correct it by hand with the **bearing** slider.

**No cloud forecast appears**

“Could not fetch the forecast” means you are offline or MET Norway is having trouble; “No forecast for tonight yet” means that night has not been published yet. Neither affects the chart.

**Notifications do not arrive**

Check Settings → Astrarium → Notifications. The schedule is rebuilt when the app is opened, so if it has not been opened for a long time nothing further ahead is queued — open it once.

**The widget shows old figures, or “As of …”**

Widgets and Siri show what was worked out when the app was last opened. Open the app once to refresh them.

# 13. Glossary

| Term | Meaning |
|---|---|
| Magnitude | Brightness; smaller is brighter, and one magnitude is a factor of about 2.5 |
| Azimuth / altitude | Direction round the horizon (north 0°, east 90°) and angle above it |
| RA / Dec | Coordinates fixed to the sky — the system star catalogues use |
| Local sidereal time | The right ascension currently on the meridian |
| Universal Time | Time at longitude 0° |
| Moon age | Days since new Moon; 0 is new, about 14.8 is full |
| Illumination | The fraction of the Moon’s disc that is lit |
| Airmass | Thickness of atmosphere, 1 at the zenith; the larger, the worse the seeing |
| ZHR | Meteors per hour with the radiant overhead and a 6.5-magnitude sky |
| Radiant | The point the meteors of a shower appear to come from |
| TLE | A satellite’s orbital elements; they go stale in a few days and need refreshing |
| Kp | The index of geomagnetic disturbance (0–9); the higher it is, the further the auroral oval reaches towards the equator |
| Geomagnetic latitude | Latitude measured from the geomagnetic pole — what the aurora actually follows |

# 14. Credits and contact

Every source Astrarium Pro computes from is listed in the app under **Info** → **Credits and sources**. The principal ones are:

- Planetary ephemeris: JPL DE421
- Stars: the Yale Bright Star Catalogue (9,096 stars)
- Deep sky: the Messier and Caldwell catalogues; photographs from Wikimedia Commons (freely licensed images only)
- Satellites: Celestrak (TLEs), SGP4/SDP4
- Space weather: NASA DONKI (solar wind, geomagnetic storms), NOAA SWPC OVATION (aurora probability)
- Deep-sky star catalogue: UCAC4 (2.34 million stars — V is measured by APASS; R is estimated from V and B-V, not measured)
- Comets and asteroids: Minor Planet Center, JPL Small-Body Database
- Earth orientation: IERS finals2000A
- Cloud forecast: MET Norway (CC BY 4.0)
- Light-pollution estimate: NASA Earth Observatory (Black Marble 2016)

Author: Shinsuke Abe a.k.a. Avell
Home page: https://avellsky.github.io/Astrarium/
Contact: https://x.com/AvellSky

Astrarium Pro © Avellsky